

Plant Pests, Diseases & Disorders

TAFE Horticulture - Class Study Notes

About these notes: Compiled from class whiteboard and presentation photographs as a study resource. Not an official TAFE SA publication. Check current assessment instructions, authoritative references, product labels and SDS information where required.

1. Plant Diagnosis

Diagnosis pathway

- Identify the plant.
- Identify the symptoms.
- Identify the issue using suitable diagnostic resources.

Site and plant factors to investigate

- Positioning: aspect, level/type of shade, slope, exposure, coastal influence and microclimates such as near fences.
- Size of the garden bed and age of the plant.
- Soil type and characteristics.
- Nutrition and irrigation.
- Water stress: consider both too little and too much water.
- Root condition and possible root pathogens.
- Environmental stress such as hot weather.

2. Symptoms and Disorders

Leaf symptoms

- Chlorosis: note whether it occurs on old or new leaves.
- Record where symptoms occur: margins, tips, base or interveinal areas.
- Red colouration may be diagnostically relevant.
- Necrosis means tissue death.

Other observations

- Fruit development.
- Flower development.
- Root development.
- Soil pH.
- Availability of water.
- Nutrient toxicities.

Nutrient movement: Old versus new leaf discolouration can help distinguish mobile from immobile nutrient deficiencies.

N-P-K purpose

- N - vegetative growth.
- P - root development; fruit and flower development.
- K - fruit and flower development.

Non-living disorders noted in class

- Mutation.
- Sunburn.
- Nutrient deficiencies.
- Chemical burn.
- Water-stress symptoms.

3. Pests and Diseases

Pests - animal groups

- Vertebrates.
- Insects.
- Mites.
- Snails and slugs.
- Nematodes.

Disease groups

- Bacteria.
- Viruses.
- Fungi.
- Phytoplasma.

Fungal

- Multicellular; may form colonies.
- Heterotrophic.
- Hyphae are filaments.
- Fruiting bodies produce spores.
- Examples noted included toadstools/mushrooms and rust as a mass of spores.

Bacterial

- Single cells.
- Microscopic.
- Reproduce asexually.
- May live as colonies.

Viral

- Recorded in class as "not alive".
- Contain either DNA or RNA as genetic material.
- A virus introduces genetic material and reprograms a host cell.

4. Phytophthora

Conditions favouring Phytophthora

- Wet soil.
- Poor drainage.
- Poor horticultural hygiene.
- Susceptible plantings.
- Alkaline pH.
- Calcium-deficient soils.
- Warm soil.
- Spread through contaminated water and contaminated soil.

Control notes from class

- Plant resistant species.
- Mulch.
- Test soil pH and adjust where appropriate.
- Test soil texture and structure and adjust where appropriate.
- Prune/remove dead or dying plants.
- Use signage.
- Organic fertiliser - Bush Tucker was noted in class.
- Trichoshield root dip was noted in class.

5. Integrated Pest Management (IPM)

Class topics included pest review, monitoring and thresholds, biological control and practical disorder planning. Identify the problem, monitor pest activity and plant damage, consider thresholds, and select suitable control measures rather than automatically relying on chemical control.

6. Pesticides and Fungicides

Common pesticides noted in class

- Oils - e.g. Eco Oil / white oil.
- Horticultural soaps - potassium salts of fatty acids.
- Pyrethrum / pyrethrins / pyrethroids; examples recorded included tau-fluvalinate, permethrin, allethrin and beta-cyfluthrin.
- Imidacloprid.
- Success (spinetoram).

- Carbaryl.
- Rotenone.
- Neem oil.

Other products/materials noted

- Mancozeb Plus.
- Wettable sulfur.
- Lime sulfur.

Common fungicides noted

- Eco-fungicide (potassium bicarbonate).
- Phosphorous acid.
- Copper products - e.g. liquid copper and copper oxychloride.

Important: Always follow the current registered product label, SDS, workplace procedures and applicable legal requirements before pesticide use. These class notes do not replace the product label.

7. Diagnostic Resources

- Reference books.
- Reliable websites.
- Image searches as an initial comparison aid.
- Garden centres.
- Botanic Gardens experts.
- TAFE SA.
- Universities.
- SARDI.
- Plant scientists.
- Land management boards / Landscape SA resources.
- Growers.
- PIRSA.

8. Practical Assessment - Hazard Prompts

Hazard	Possible risk	Controls to consider
Slips, trips and falls	Fall / injury	Inspect site; remove or isolate hazards; suitable footwear.
Biological organisms	Bites / stings	Inspect area; avoid nests; appropriate PPE and procedures.
Sharp tools	Cuts / punctures	Inspect and use tools correctly; keep hands clear; store safely.
Manual handling	Strain / injury	Correct lifting technique; reduce loads; seek assistance where needed.

Hazard	Possible risk	Controls to consider
UV exposure	Sun / heat exposure	Shade where possible; protective clothing, hat and sunscreen.
Chemical use	Exposure, drift or non-target effects	Follow label and SDS; specified PPE; correct application; suitable weather.

Monitoring example: Regular visual inspection. **Identification technique:** examine potential pests and compare them with an appropriate identification guide or other reliable diagnostic resource.